SYNOPSIS REPORT

ON

[COURSE CONTENT VERSION CONTROL SYSTEM (VCS)]
VERSION CONTROL SYSTEM, CSE2480

SCHOOL OF COMPUTER SCIENCE

IILM UNIVERSITY, GURUGRAM

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School of Computer Science

IILM University, Gurugram

Project Description (2026, 4th Semester)

Project Title: Course Content Version Control System (VCS)

Deployed at: nitinyi.github.io/vcs-project/

Technology Used	HTML5, CSS3 (Flexbox, Responsive Design), JavaScript
Deployment	GitHub Pages (Frontend Only, No Backend)
Version Control	Git, GitHub

1. Abstract

The current proposal involves a lightweight, front-end-only course content version control system that utilizes HTML, CSS, and JavaScript. The system fetches the history of revisions made to course material and displays the latest as well as previous versions.

Enhancement Added

The project now includes a modern, responsive, and user-friendly UI with:

- A centered login interface with image integration
- A premium gradient-based theme (pink-purple blend)
- A responsive layout optimized for both mobile and laptop screens
- An improved dashboard interface with sidebar navigation and structured file display

The project runs entirely on GitHub Pages and does not require a backend.

2. Introduction

This project is developed under the domain of Web Applications and Management Systems, focusing on creating a user-friendly system for managing academic course content. In today's digital learning environment, educational materials are constantly updated with new information, corrections, and improvements. However, managing these updates efficiently remains a challenge, especially when traditional file-sharing methods are used. These methods do not provide version history, leading to confusion about which file is the latest or what changes have been made over time.

The significance of this domain has grown rapidly as education becomes more technology-driven. Students and faculty rely heavily on digital platforms, and there is a strong need for systems that ensure clarity, organization, and easy access to updated content. A Version Control System (VCS) tailored for course materials helps solve this problem by allowing users to track changes, maintain previous versions, and manage content in a structured way. This improves collaboration, reduces errors, and enhances overall efficiency in academic workflows.

The project introduces a Course Content Version Control System (VCS) built entirely using front-end technologies. This makes the system lightweight, easy to deploy, and accessible directly through a web browser without requiring any server-side setup. The main objective is to provide a simple yet effective platform where users can manage multiple versions of course content, compare updates, and maintain a clear record of changes.

In addition to functionality, the project strongly focuses on user experience and interface design. Modern applications are expected to be visually clean and easy to navigate, which directly impacts usability. To meet these expectations, the system includes a well-designed login page, a clean and structured dashboard, and responsive layouts that adapt to different screen sizes. Proper spacing, readability, and intuitive navigation have been prioritized to ensure that users can interact with the system comfortably.

Overall, this project not only addresses the technical challenge of version control in an academic setting but also delivers a modern and accessible interface. It combines practicality with good design, making it useful for both students and educators while aligning with current trends in web application development.

3.Problem statement

This project is developed under the domain of Web Applications and Management Systems, focusing on creating a user-friendly system for managing academic course content. In today's digital learning environment, educational materials are constantly updated with new information, corrections, and improvements. However, managing these updates efficiently remains a challenge, especially when traditional file-sharing methods are used. These methods do not provide version history, leading to confusion about which file is the latest or what changes have been made over time.

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4. Objectives

- Build a front-end-only version control system for course content
- Allow teachers to upload course files with commit messages
- Allow students to view and download the latest or previous versions of files
- Integrate GitHub API to fetch real version history
- Design a modern and responsive user interface
- Create a centered login page with visual elements (image + card layout)
- Ensure cross-device compatibility (mobile + laptop)
- Improve user experience through better spacing, colors, and layout
- Build a dashboard with sidebar navigation and file management UI

5. Methodology

5.1 System Architecture

The system is entirely front-end based. It uses the GitHub API to fetch commit history and file versions stored in a public GitHub repository.

UI Layer Enhancement:

- Login page redesigned using flexbox centering
- Use of card-based layout
- Gradient theme implementation
- Responsive media queries for different screen sizes

5.2 Workflow

Teacher Workflow:

- Teacher signs in → selects role → uploads files → enters commit message → files are updated

Student Workflow:

- Student signs in → views files → downloads latest or previous versions

UI Enhancement Workflow:

- User lands on login page → interacts with clean UI → logs in → redirected to structured dashboard → performs actions easily

5.3 Technology & Tools

Front End	HTML5, CSS3 (Flexbox, Responsive Design), JavaScript
UI Design	Gradient themes, card layouts, spacing system
Deployment	GitHub Pages
Version Control	Git, GitHub

6. User Interface Improvements

The following UI enhancements were implemented to improve overall usability, aesthetics, and cross-device compatibility:

6.1 Login Page Improvements

- Centered login card using flexbox
- Split layout with image and form
- Gradient login button (pink-purple theme)
- Improved spacing and typography
- Responsive behavior for mobile and desktop

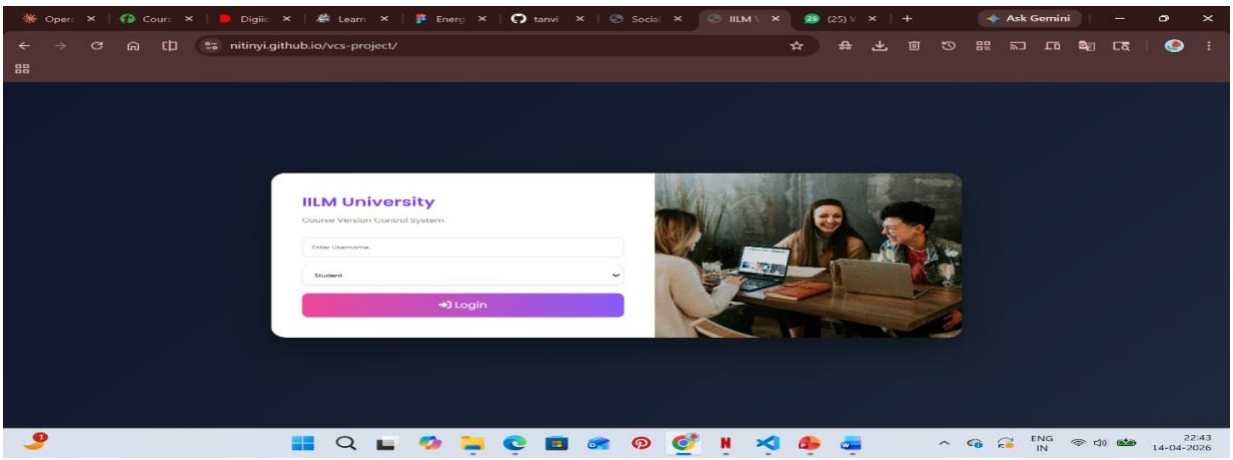
6.2 Dashboard Improvements

- Sidebar navigation added
- User role display (student/teacher)
- Styled file cards instead of plain text
- Upload section redesigned
- Improved color contrast and readability

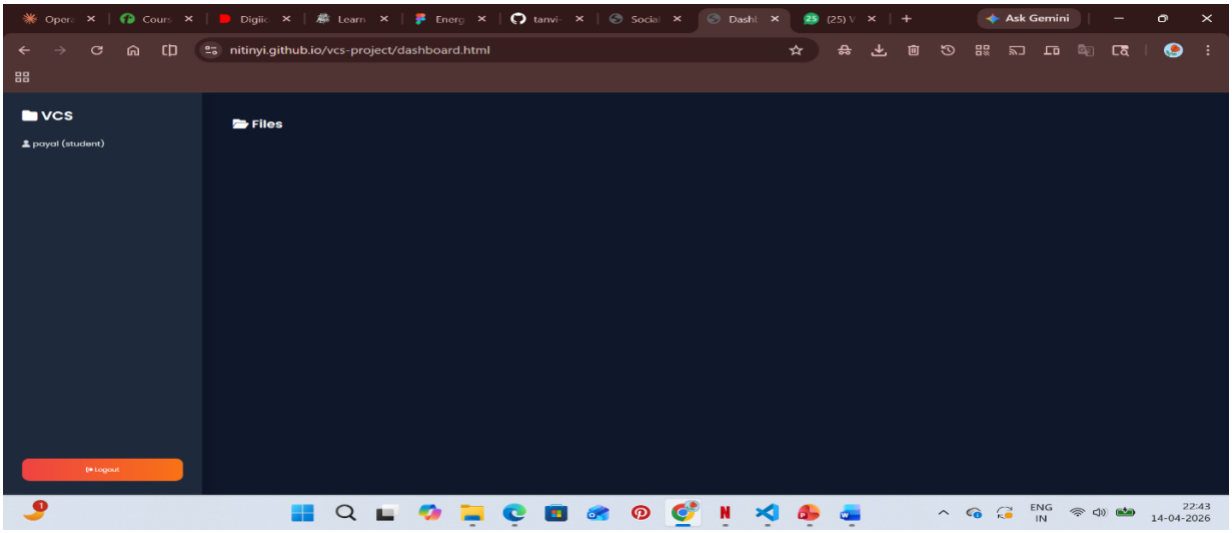
6.3 Responsiveness Fixes

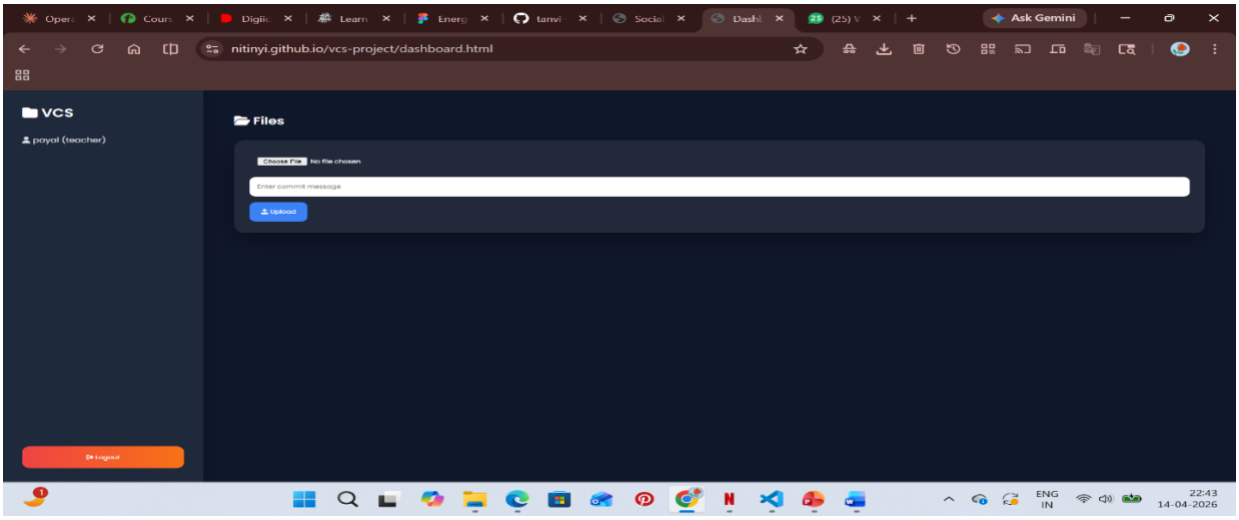
- Fixed issue where UI appeared too large on laptops
- Added proper width constraints
- Ensured layout adapts across screen sizes

LOGIN PAGE



DASHBOARD





7. System Requirements

7.1 Hardware Requirements

- Any device with a modern web browser (PC, Laptop, Mobile, Tablet)
- Internet connection for accessing GitHub Pages and API

7.2 Software Requirements

- Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)
- GitHub Account (for deployment)
- Code Editor: Visual Studio Code (for development)
- Git (for version control)

8. Project Schedule

Week	Task / Milestone
Week 1	Requirement gathering and problem analysis
Week 2	System design and architecture planning
Week 3	Frontend development – login page and role selection
Week 4	Dashboard UI design and file display system
Week 5	GitHub API integration and version fetching logic
Week 6	UI enhancements – responsiveness, gradients, card layouts
Week 7	Testing across devices and bug fixes
Week 8	GitHub Pages deployment & final demo (includes UI enhancements and responsiveness fixes)

9. References

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2. MDN Web Docs – HTML, CSS, JavaScript – https://developer.mozilla.org
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4. Version Control Systems – Wikipedia – https://en.wikipedia.org/wiki/Version_control
5. Pro Git Book – Scott Chacon & Ben Straub – https://git-scm.com/book
6. UI/UX Design Inspiration – Dribbble / Behance
7. CSS Flexbox Guide – https://css-tricks.com/snippets/css/a-guide-to-flexbox/

Synopsis Draft verified

Project Guide **Panelist1** **Panelist 2**

(Name & Sign) **(School of Computer Science)**